

Programming For Data Analytics 1

Assignment 1 (Part 1)

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Title: Using Linear Regression Analysis to develop a Sliding Scale of Charges for a Medical Aid Scheme based on Lifestyle Features and geographical location

Dataset available at: [Medical Cost Personal Datasets](#)

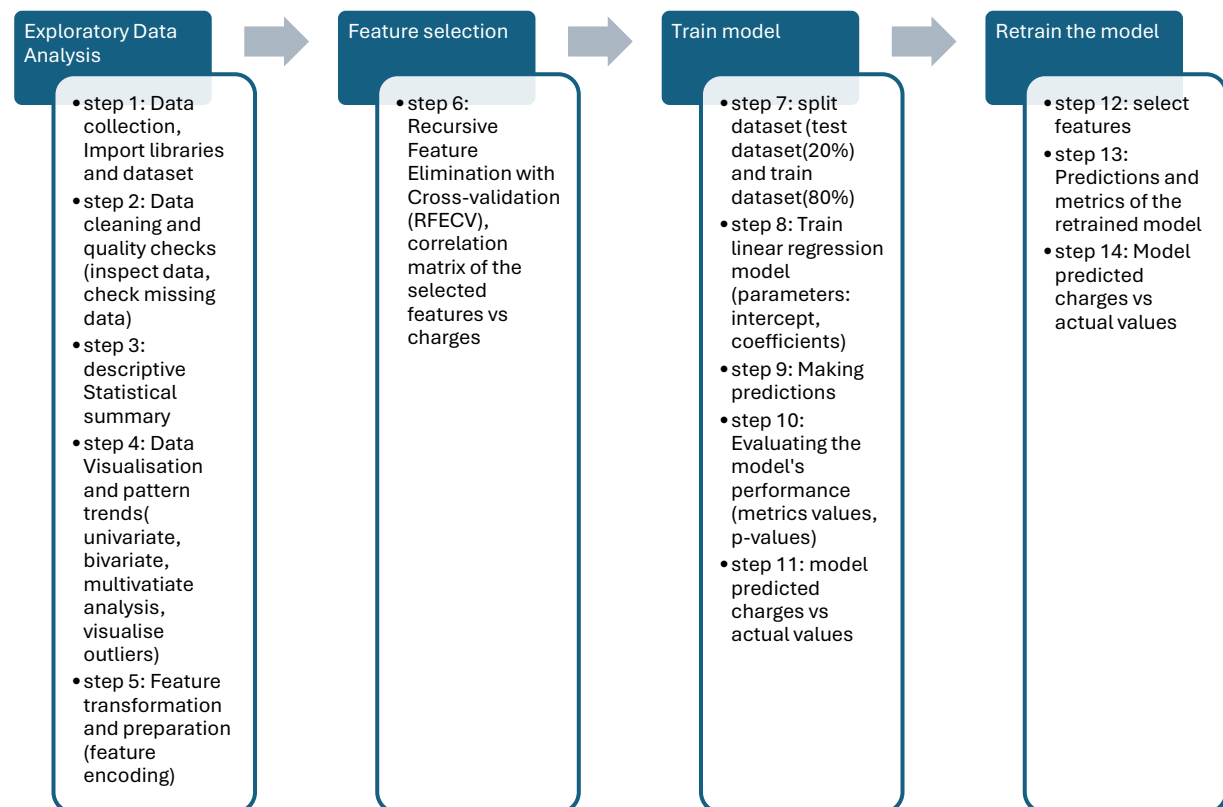


Figure 1: report planning

Introduction:

This report assesses the application of a Linear Regression model that predicts medical insurance costs based on client's lifestyle and demographic data. We want to use the provided dataset to develop a "sliding scale" of medical aid plan expenses. The aim is to create a "sliding scale" of charges for a medical aid plan based on features that have a high influence on charges. Exploratory Data Analysis (EDA) identifies the major drivers of medical expenses, resulting in an accurate and equitable pricing tool.

Why data is suitable for linear regression model?

This data is suitable for linear regression because, regression model is designed to predict a continuous numerical value, 'charges' column has numerical values not categorical values.

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Importing Libraries and Data

The initial stage in exploratory data analysis is to collect data from a variety of sources, including databases, spreadsheets, and Kaggle. Following collection, the necessary libraries are imported, and the data is then loaded into a tool or programming environment, such as Python, where it may be viewed and inspected (Simplilearn, n.d.). The Us-dataset was collected from Kaggle.

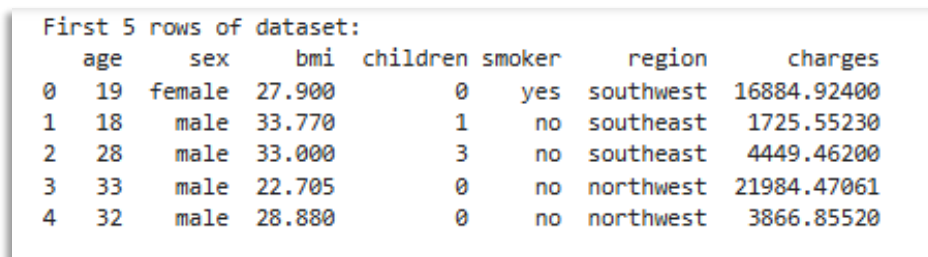
IMPORT LIBRARIES

To create a linear regression model that provides a sliding scale of charges according to the age, sex, bmi, number of children, smoking status, and geographical region of the medical aid scheme's clients perform exploratory data analysis on medical insurance data. The first step was to import all the necessary libraries and inspecting

the data. Pandas and NumPy are used to transform, clean, and arrange data into quick numerical arrays. Matplotlib and Seaborn generate plots or visual presentations that allow you to observe patterns visually. Finally, Scikit-Learn (sklearn) includes tools for splitting your data for testing, running the Linear Regression method, and using metrics such as R-squared to assess how effectively your model predicts the outcomes.

LOAD DATASET

The dataset was imported. The data was read with pandas, and `df.head()` displays the top 5 observations from the dataset. This is useful since it shows that the dataset was loaded into Python.



```
First 5 rows of dataset:
  age  sex  bmi  children  smoker  region  charges
0   19 female 27.900         0    yes southwest 16884.92400
1   18  male 33.770         1     no  southeast  1725.55230
2   28  male 33.000         3     no  southeast  4449.46200
3   33  male 22.705         0     no northwest 21984.47061
4   32  male 28.880         0     no northwest  3866.85520
```

Figure 2: Top 5 observations of the dataset

Exploratory Data Analysis (EDA)

Exploratory data analysis is a data analytics approach that uses visual means to comprehend the depth and properties of data, and it is performed at the beginning of a project. It helps to summarize the data and reveal insights before moving on to more complex data analysis approaches (Simplilearn, n.d.).

In machine learning, EDA analysis prepares data prior to model development. It aids in the detection of outliers, class imbalances, and patterns that may influence the outcome. Analysts investigate how input features connect to the target variable, which enhances data cleansing, feature selection, and model performance (Simplilearn, n.d.).

DATA INSPECTION

The data was examined to ensure suitability for predictive modelling. The information regarding the data and its nature, including missing values (Null), the number of records in each column, and memory use, were examined. The Medical insurance dataset has seven columns labelled age, sex, BMI, children, smoker, region, and costs. It contains 1338 entries. The age column represents the client's age, the sex column represents the client's gender, bmi represents the client's body mass index, the children column represents the number of children/dependents the client has, the smoker column tells us the client's smoking status, the region column tells us the client's geographical region, and finally the charges column represents the medial insurance costs. These seven columns include four columns of numerical data: age, BMI, children, and charges. There are three columns with categorical data: sex, smoker, and region. The Datatype: age column contains 64 integers; sex column contains object datatype; bmi column contains 64 floats; children column contains 64 integers; smoker column contains object datatype; region column contains object datatype; charges column contains 64 floats.

```
# generate the information/ structure about the dataset
print("\nDataset Info:")
print(df.info())
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1338 non-null   int64
1   sex         1338 non-null   object
2   bmi         1338 non-null   float64
3   children    1338 non-null   int64
4   smoker      1338 non-null   object
5   region      1338 non-null   object
6   charges     1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
None
```

Figure 3: Dataset Info

DATA CLEANING

This process entails correcting or deleting errors, addressing missing values, and resolving discrepancies. Analysts also look for duplication and formatting issues to verify that the dataset is suitable for further analysis.

Missing Values

For this dataset we checked for missing values. `IsNull().sum()` checks the total of missing values (Null) in each column. Missing values might have a negative impact on the model's performance, so they should be eliminated or filled. The results showed that the data had no missing values. There were no severe data quality issues found; hence the dataset was declared suitable for analysis.

```
# For data cleaning check missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

Missing Values:

age	0
sex	0
bmi	0
children	0
smoker	0
region	0
charges	0
dtype:	int64

Dataset Shape: (1338, 7)

Figure 5: Table showing total number of missing values

DESCRIPTIVE STATISTICAL SUMMARY

Following cleaning, basic summary statistics like mean, median, mode, range, and standard deviation should be generated. These measurements provide a snapshot of

the data distribution, assisting in the identification of odd values or probable data-entry issues (Simplilearn, n.d.).

Then the data's summary statistic was examined. This can be accomplished by calling `describe()`, which returns a statistical summary of numerical datatypes. To produce a statistical overview of all data, including object or categorical values, use `describe(include='all')`. In this scenario, `describe()` was employed. The studies revealed that ages ranged from 18 to 64 years, with 39 being the average age. BMI ranges from 15.96 to 53.13, with an average of 30.66 falling into the obese group according to standard BMI standards. The medical insurance charges range from \$1121 to 63770.

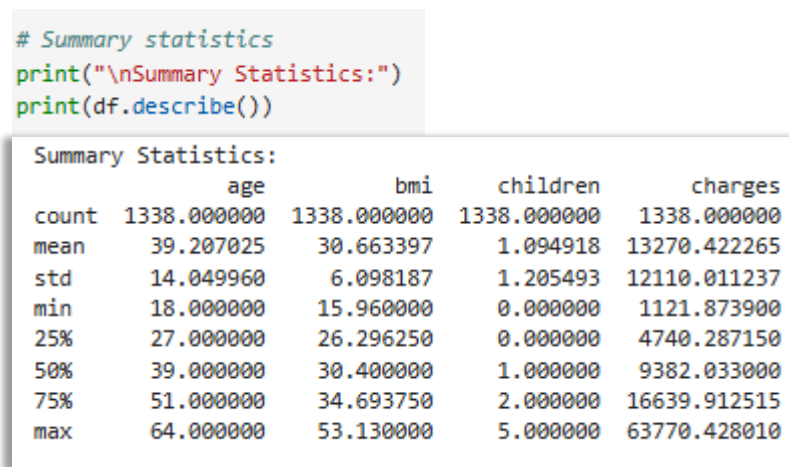


Figure 4: Summary of statistics

DATA VISUALISATION

Univariate Analysis

Univariate analysis can be used for both categorical and numerical data. Numerical variables can be visualized using histograms, box plots, density plots, and so on (Malamahadevan, 2025). Matplotlib and Seaborn were used to visualize all our graphic plots.

Because our objective variable is "charges", a graph depicting the distribution of charges was created. A distribution graph determines whether data is "skewed" or "normal". Seaborn was used to generate a histogram of the charges column in the data frame (df). To display the data distribution, set `kde=True`, which adds a kernel

density estimate plot (a smooth curve) on top of the histogram. The independent variable (x-axis) was labelled as charges, followed by the dependent variable (y-axis) labelled as frequency. The resulting plot indicated that the graph is skewed to the right (positively). This means that most people have relatively low charges (the left side of the graph is tall), while only a few have extremely large charges (the right tail stretches out). The tallest bar is around the charge range of 0 to 10,000. This implies that the most typical charge values fall within that range, with many claims for amounts less than \$10,000. As charges exceed 10,000, the bars become shorter, indicating that fewer observations contain those higher values. By the time you reach 50,000-60,000, the frequency is nearly zero. The data reveals that insurance claims are typically small, with rare huge outliers. Models and analyses should account for this skewness.

```
#distribution of charges
#create a new matplotlib figure with a size of 8 inches wide and 5 inches tall
plt.figure(figsize=(8,5))
#use sns to draw histogram of the charges column, kernel density estimate (kde)=True adds a smooth curve over the histogram to show data distribution.
sns.histplot(df['charges'],kde=True)
plt.title("Figure 1: Distribution of Medical Insurance Charges")
plt.xlabel("charges")
plt.ylabel("Frequency")
plt.show()
```

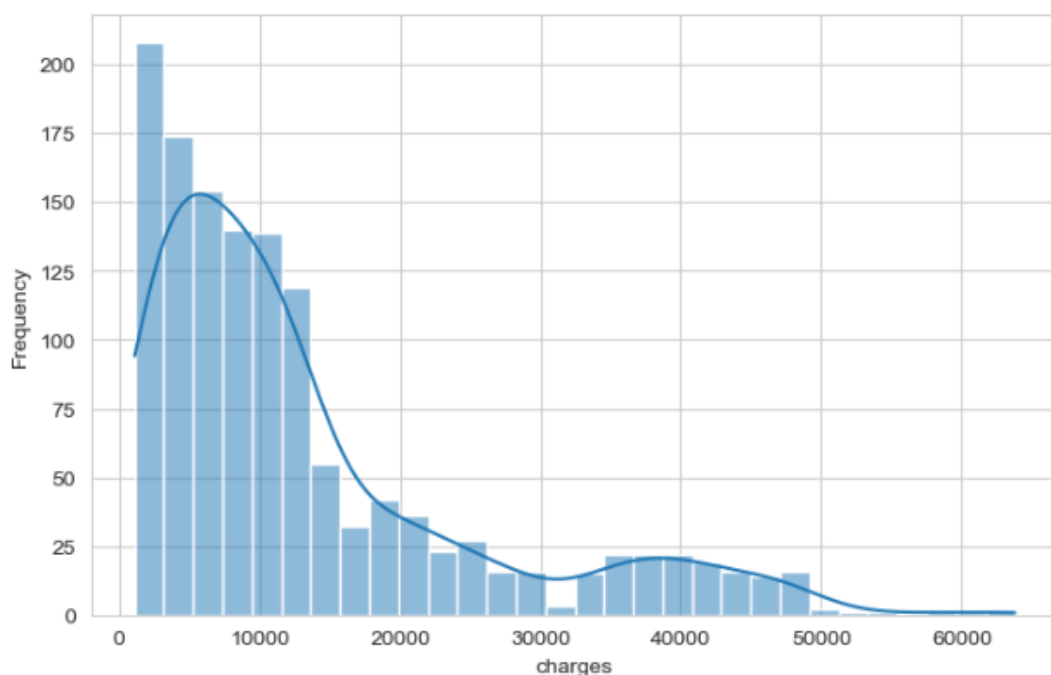


Figure 6: Distribution of medical Insurance Charges

A boxplot of charges was created to show the outliers. The boxplot of charges reveals that clients are usually charged between \$1000 and 18000, with a median of around

\$8000, while a few numbers extend far to the right. Outliers were found in the charges variable. However, these were kept since they represent real-world high-cost insurance instances and are necessary for developing a realistic prediction model.

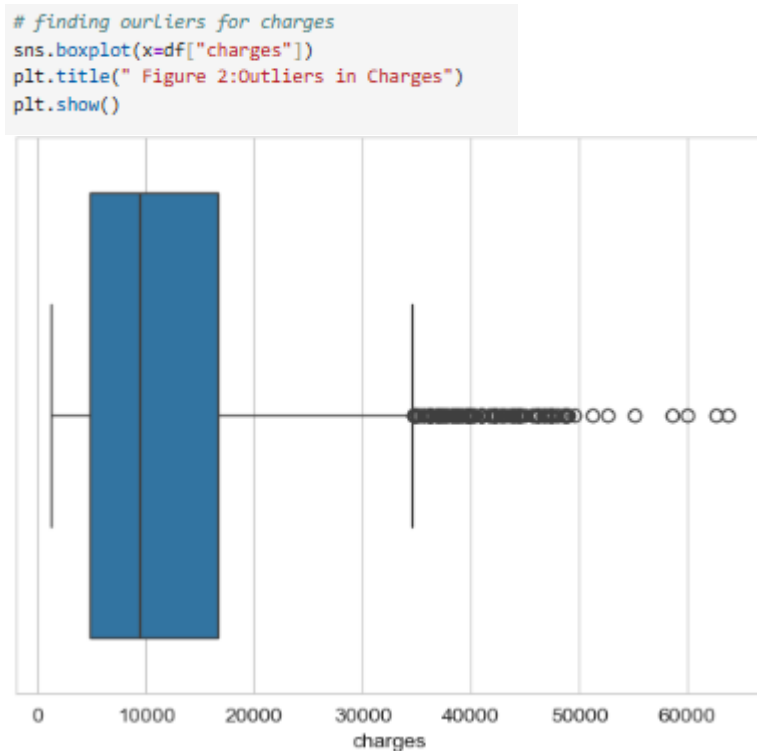


Figure 7: Boxplot of charges

Bivariate analysis: Numeric-Numerical analysis

Bivariate analysis assists in understanding how variables are related to one another, as well as the relationship between dependent and independent variables in the dataset (Malamahadevan 2025)

A pair plot was used to compare multiple variables (age, BMI, children, and charges). A scatter plot of all feature factors (age, BMI, and children) against charges was created. The plot shows that charges tend to increase with age, though the connection appears fairly linear with some variation. As bmi grows, the scatter points show an increased trend in charges; the distribution of points is rather dense between bmi 20 and 40, with charges ranging from low to high. Outliers might arise at different bmi levels. The broader distribution shows that bmi has a lesser influence at that age. The vertical strips of points on the children versus charge plot demonstrate that the number

of children alone cannot fully explain charge fluctuation; other factors may influence the charges.

```
#pair plot
# indicate the numeric columns which will be x_vars, and the target variable (charges) will be y_vars
numeric_cols=['age', 'bmi', 'children']
sns.pairplot(df, x_vars=numeric_cols, y_vars='charges', kind='scatter')
plt.suptitle('Figure 4: Numerical Features vs Medical Charges', y=1.05)
plt.show()
```

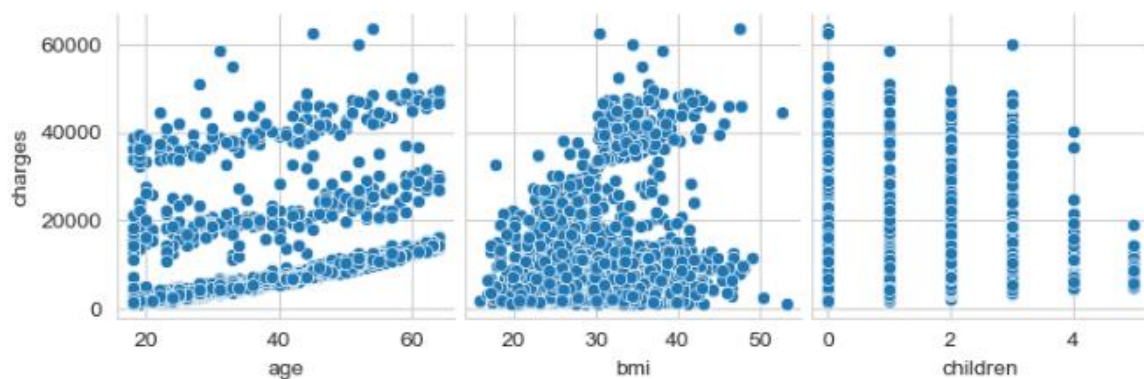


Figure 8: Numerical features vs medical charges.

Bivariate analysis: Numeric-Categorical analysis

The Numeric-Categorical analysis used group by function to divide the data into related categories. Rows with the same value in a given column are grouped. This happens so that numerical occurrences of a specific category across a column can be seen.

The median value of costs for clients with a smoking status of yes or no was determined. Then a boxplot of smokers' vs charges was generated. A box plot depicts the range of values that fall into a specific category. The boxplot below illustrates that clients who smoke have costs ranging from \$22000 to \$42000, with a median of \$34456.34845, which is higher than that of non-smokers. The fees for people who do not smoke range from \$1100 to \$1300, with a median of around \$6 000. The points outside of the box are outliers, however these were retained since they represent real-world high-cost insurance cases, which are vital for constructing a realistic predictive model.

```
#groupby the smokers to find the statistical analysis of the charges with response no & yes separately.
df.groupby('smoker')['charges'].median()

#plot the box of charges with response(smoker) no & yes.
sns.boxplot(x=df['smoker'], y=df['charges'])
plt.title('Figure 6: Boxplot of smoker with respect to charges')
plt.show()
```

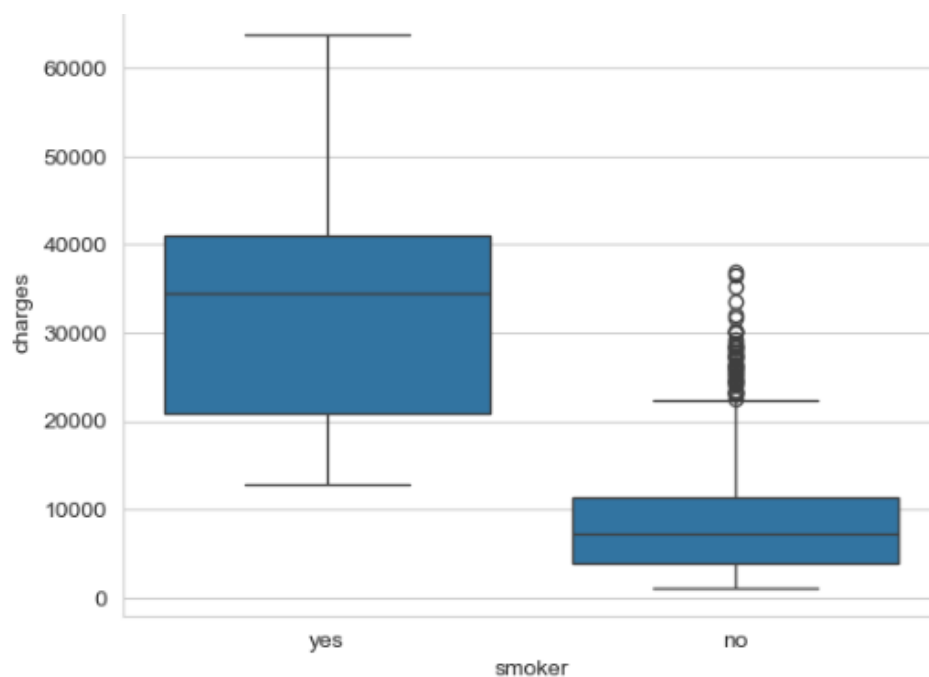


Figure 9: Boxplot of smoker with respect to charges

The median value of costs for clients with sex statuses of female or male was determined. Then a boxplot of smokers vs charges was generated. A box plot depicts the range of values that fall into a specific category. The boxplot below reveals that males are charged between \$1100 and \$20,000, with a median of \$9369.61575. The remaining values are outliers. Females are charged \$1600-15000, with a median of \$9412.96250; the remaining values are outliers, according to the boxplot. The medians and distributions of the two genders are nearly identical, indicating there is no difference. The points outside the box are outliers; however, they were kept since they represent real-world high-cost insurance instances and are necessary for developing a realistic prediction model.

```
#groupby the sex to find the statistical analysis of the charges with response male & female separately.
df.groupby('sex')['charges'].median()

#plot the box of charges with response(sex) female & male.
sns.boxplot(x=df['sex'], y=df['charges'])
plt.title('Figure 7: Boxplot of sex with respect to charges')
plt.show()
```

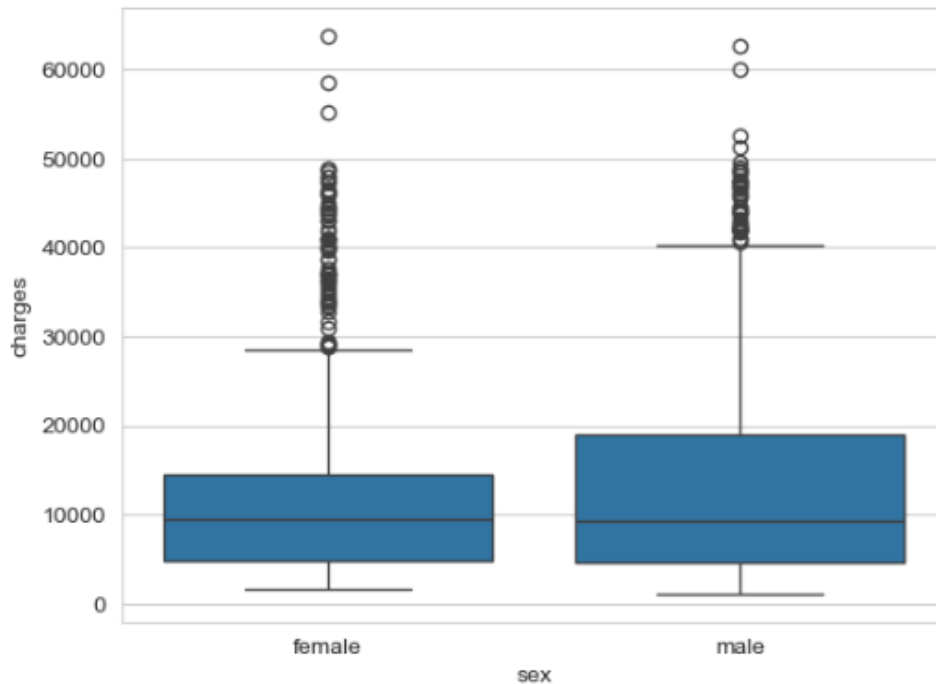


Figure 10: Boxplot of sex with respect to charges

The median value of costs in respect to the client's geographical region was determined. Then a boxplot of region versus charges was created. A box plot depicts the range of values that fall into a specific category. The boxplot below shows that some southeast region clients may be charged more because they have a higher maximum charge than the other regions; charges in the southeast region range from \$1100-19000, with a median of \$9294.131950; the other points outside the box are outliers, but the medians of all regions are nearly identical. The points outside the box are outliers; however, they were kept since they represent real-world high-cost insurance instances and are necessary for developing a realistic prediction model.

```
#groupby the region to find the statistical analysis of the charges
df.groupby('region')['charges'].median()

#plot the box of charges with the responses of region.
sns.boxplot(x=df['region'], y=df['charges'])
plt.title('Figure 8: Boxplot of region with respect to charges')
plt.show()
```

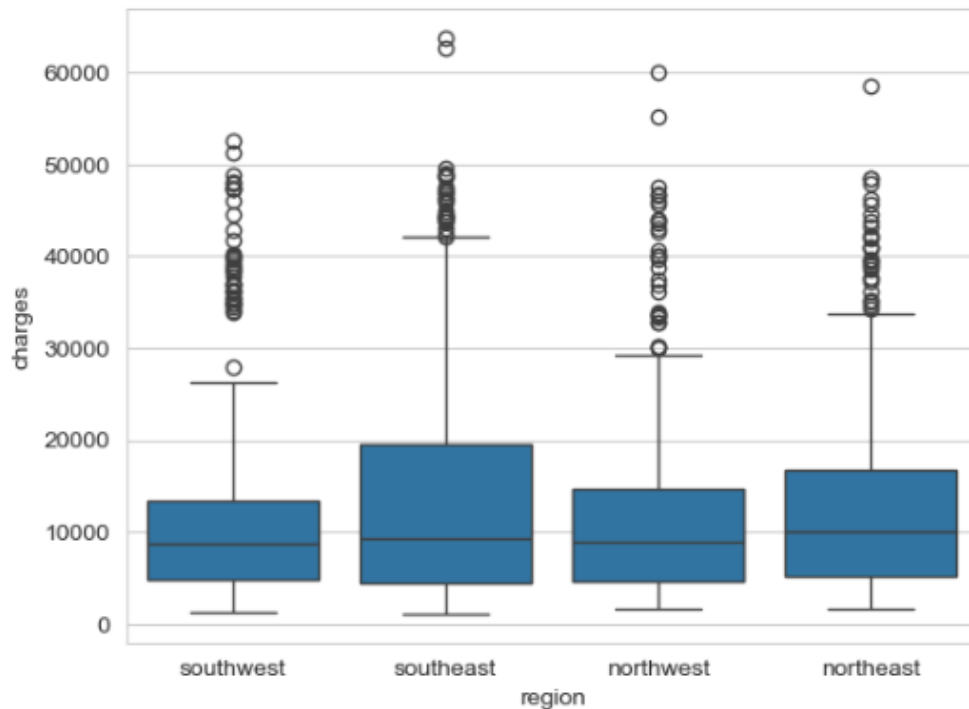


Figure 11: Boxplot of region with respect to charges

Multivariate analysis

Multivariate analysis looks at the link between more than two variables. A heat map is extensively used in Multivariate Analysis because it indicates the correlation between variables and whether they are positive or negative (Malamahadevan, 2025). A correlation matrix was utilized to determine the relationship between various variables. The correlation coefficient determines how two variables are related. The correlation matrix is shown in the figure below; in this case, we examined the relationship between charges and age, bmi, and children to determine how these factors influence charges. The association matrix shows that Charges versus Age has a diagonal value of 0.3, indicating a modest positive association; as age grows, charges tend to climb slightly. Older people tend to have slightly greater charges.

Charges versus bmi has a 0.2 correlation, indicating a mild positive relationship; higher bmi is associated with modestly high charges. The correlation coefficient between charges and children is 0.068, indicating a very weak positive relationship.

```
# create a correlation matrix
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()
```

Figure 12: Correlation matrix



Outliers

The target variable, 'Charges,' exhibited outliers. However, the outliers were not deleted because doing so would result in a model that underestimated the schemes' financial exposure for high-cost claims. By removing outliers, the clients with the riskiest lifestyles (heavy smokers, high BMI) will also have their charges removed.

FEATURE TRANSFORMATION AND PREPARATION

The final stage in EDA is to prepare the data for modelling or deeper analysis. This could include modifying variables, scaling numbers, encoding categories, or developing new features based on previously identified patterns. Well-prepared features improve model performance and accuracy (Simplilearn, n.d.).

Feature Encoding

Before selecting features, the categorical values in the columns (sex, smoker, and region) were converted to numerical values. This step was necessary to ensure that the model could effectively analyse categorical data. A one-hot encoding approach was used to convert category variables to numerical data (Singh, 2025). Drop_first=True removes the first category to avoid redundancy and multicollinearity, whereas dtype=int converts Boolean True/False to 1 and 0. Drop_first=True decreases the number of features in a linear regression model, making it simpler and faster to train (Singh, 2025).

```
Encoded Dataset Preview:
   age    bmi  children    charges  sex_male  smoker_yes  region_northwest  \
0   19  27.900         0  16884.92400         0           1              0
1   18  33.770         1   1725.55230         1           0              0
2   28  33.000         3   4449.46200         1           0              0
3   33  22.705         0  21984.47061         1           0              1
4   32  28.880         0   3866.85520         1           0              1

   region_southeast  region_southwest
0                0                1
1                1                0
2                1                0
3                0                0
4                0                0
```

Figure 13: Catagorical features converted to numerical data

Feature scaling

Feature scaling was not used in this investigation since Linear Regression models are not sensitive to changes in feature scales (GeeksforGeeks, 2025). As a result, the model can still perform well without normalisation.

FEATURE SELECTION

To choose features, the Recursive Feature Elimination with Cross-validation (RFECV) approach was used. RFECV runs the model with all features and then performs cross-validation to determine which number of features yields the maximum accuracy score (Magedmallek, n.d.). This improved model efficiency and reduced the possibility of overfitting by removing irrelevant or unnecessary variables. RFECV and the linear regression model were imported from sklearn. When generating the feature matrix X, the target column (charges) was removed from the encoded data frame, and the variable we wish to predict was assigned the name "charges". The optimal number of features identified by RFECV was printed, and selected feature names from X.columns were extracted with the rfecv.support_mask. The output indicates that 8 attributes were chosen as optimal. The selected features are age, BMI, children, sex_male, smoker_yes, region_northwest, region_souteast, and region_southwest.

```
from sklearn.feature_selection import RFECV
from sklearn.linear_model import LinearRegression

# drop "charges" column
X = df_encoded.drop("charges", axis=1)
y = df_encoded["charges"]

#initialise the model
model=LinearRegression()

# Use RFECV, with minimum features to select as 1, and a cross validation of 10, meaning 10 different versions of the data will be tested
rfecv=RFECV(estimator=model, step=1, cv=10, scoring='neg_mean_squared_error')
fit=rfecv.fit(X,y)

# select number of features with highest accuracy score
print(f"optimal number of features:{rfecv.n_features_}")
selected_features= X.columns[rfecv.support_]
print("Automatically selected features;",selected_features)

optimal number of features:8
Automatically selected features; Index(['age', 'bmi', 'children', 'sex_male', 'smoker_yes', 'region_northwest',
    'region_souteast', 'region_southwest'],
    dtype='object')
```

Figure 14: Feature selection

Correlation matrix of selected features vs charges

A correlation matrix for feature selection was created. It aids in discovering significant features by assessing their relevance to the target variable and deleting superfluous features. Because the features had previously been picked using the RFECV

approach, this correlation method was added to assist us see the statistical link between charges and selected features. The correlation matrix shows that Charges versus Age has a diagonal value of 0.3, indicating a modest positive association; as age grows, charges tend to climb slightly. Older people tend to have slightly greater charges. Charges versus bmi has a 0.2 correlation, indicating a mild positive relationship; higher bmi is associated with modestly high charges. The correlation coefficient between charges and children is 0.068, indicating a very weak positive relationship. Charges and sex_male has 0,057, indicating a very weak positive correlation; meaning being male has little to no influence on charges. Charges versus smoker_yes has a 0.79 correlation, which is a high positive correlation. Smokers face higher expenses than non-smokers. Charges versus region_northwest has -0.04, which is a very minor negative correlation, living in the northwest lowers charges. Region_southeast has a 0.074, indicating a very weak positive correlation; the southeast area is associated with mildly high charges. region_southwest has -0.043, which is a very modest negative correlation; residing in the southwest reduces charges. As a result, smokers have a significant influence on charges.

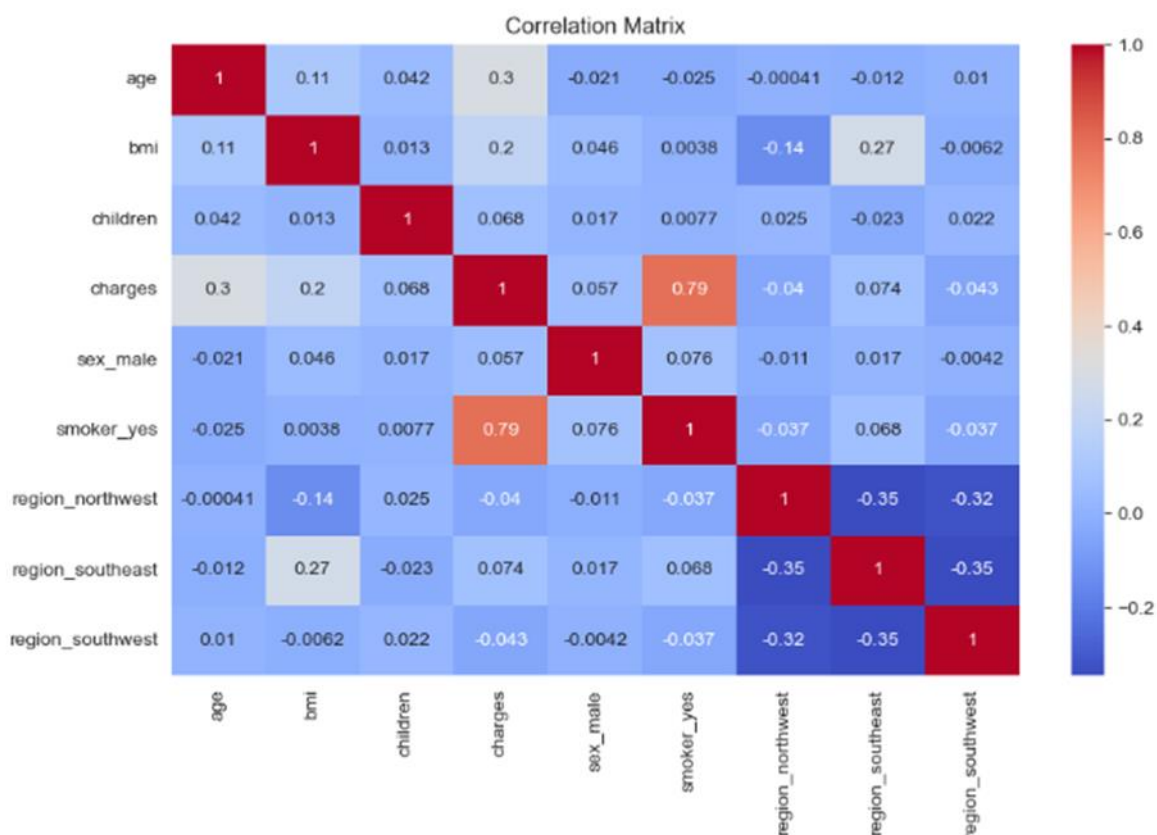


Figure 15: correlation matrix of selected features and charges

TRAIN MODEL

SPLIT DATASET

The dataset was separated into training and testing sets based on an 80/20 split. The split guarantees that the model has enough data to learn the relationship between charges and selected features, while also saving a 20% unseen dataset to objectively assess the model's forecast performance. The training set was used to develop the model, while the testing set was used to assess its performance on previously unseen data. This method also assures that the model generalizes well while lowering the danger of overfitting.

Table 1: training and testing dataset sizes

Training set size	Testing set size
(1070, 8)	(268, 8)

Figure 16: Split Dataset

TRAIN LINEAR REGRESSION MODEL

A Linear Regression model was created to forecast medical insurance charges depending on the features chosen. Linear regression was chosen because it is simple, interpretable, and successful for continuous target variables. The parameters used for this regression model are slope (coefficients) and intercept (β_0), which define the best-fit line by minimizing the sum of squared residuals. β_0 represents the intercept (-11931.219050326685), while $\beta_1, \beta_2, \dots, \beta_p$ represent the coefficients for each feature x_i .

The input features are $x_1, x_2, \dots, x_p$. (Error/residual) refers to the difference between the projected and actual costs.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$$

Table 2: intercept and coefficient values

Feature	Values
Intercept	β_0
	-11,931.2191
Coefficients	$\beta_1, \beta_2, \dots, \beta_7$
age	256.9757
bmi	337.0926
children	425.2788
sex_male	-18.5917
smoker_yes	23,651.1289
region_northwest	-370.6773
region_southeast	-657.8643
region_southwest	-809.7994

Figure 17: values for model parameters

MAKING PREDICTIONS

The trained linear regression model was used to make 10 initial predictions on the testing dataset. These forecasts show the projected medical insurance costs based on the input features.

```
First 10 Predictions:
[ 8969.55027444  7068.74744287 36858.41091155  9454.67850053
 26973.17345656 10864.11316424  170.28084136 16903.45028662
 1092.43093614 11218.34318352]
```

Figure 18: model predictions

EVALUATION OF THE MODEL'S PERFORMANCE

The model's performance was tested using the following metrics:

- Mean Absolute Error (MAE) measures the average magnitude of prediction errors.

- Root Mean Squared Error (RMSE) penalizes greater errors and measures model accuracy.
- R-squared (R^2) indicates how much of the target variable's variance is explained by the model.

The findings below show an MAE of 4181,194, indicating that the model's predictions are around 4181,194 units off on average. It prioritizes both little and major faults equally. The RMSE is 5796, 28, which is significantly greater than the MAE. This gap suggests that the model is generating significant errors on a few key data points. If the model made modest consistent mistakes, the MAE and RMSE would be much closer. The R squared score is 0.7836 (78.4%). This is a strong result, indicating that the model can explain 78% of the link between the specified features and charges. 22% of the variation in charges remains unexplained, possibly due to random noise or features not incorporated in the model. P-values were computed for each characteristic. Variables with $p < 0.05$ were identified as important drivers of medical costs, leading to their inclusion in the final premium adjustment model. Age, BMI, children, and smoker_yes are highly significant ($p < 0.05$), indicating they are reliable predictors of cost. Being a smoker has the greatest influence, adding around \$23,650 to the predicted charge. In contrast, the sex_male and all region variables have p-values significantly greater than 0.05, indicating that they have no statistically significant effect on prices in this dataset. This demonstrates that lifestyle and age are the key drivers of your "sliding scale," whereas geography and gender are less relevant for this particular medical aid system.

Table 3: Metrics values

Metrics	Value
Mean Absolute Error (MAE)	4,181.1945
Mean Squared Error (MSE)	33,596,915.8514
Root Mean Squared Error (RMSE)	5,796.2847

R ² Score	0.7836
----------------------	--------

```

Feature P-Values:
smoker_yes      9.774475e-286
age             6.310917e-70
const          1.815555e-25
bmi            4.109401e-24
children       6.063516e-03
dtype: float64

Insignificant Features (p > 0.05):
sex_male        0.960591
region_northwest 0.490070
region_southeast 0.223214
region_southwest 0.130563

```

Figure 19: Evaluation of model performance using metrics values and p-values

MODEL PREDICTED CHARGES VS ACTUAL VALUES

A comparison of actual charges and predicted charges was provided, revealing that the model's forecasts are mixed. Some forecasts were quite near to the true values, while others showed significant differences, with the model either underestimating or overestimating the actual quantities. The scatterplot graph below shows that the model accurately predicts lower insurance costs but fails when charges rise, persistently underestimating high-cost claims. Points at the upper end of the x-axis are far below the red "perfect prediction" line. The presence of three different clusters indicates that other key features were not adequately captured, resulting in a lower R square score and more total error.

```

Actual vs Predicted:
      Actual Charges  Predicted Charges
764      9095.06825      8969.550274
887      5272.17580      7068.747443
890     29330.98315     36858.410912
1293      9301.89355      9454.678501
259     33750.29180     26973.173457

```

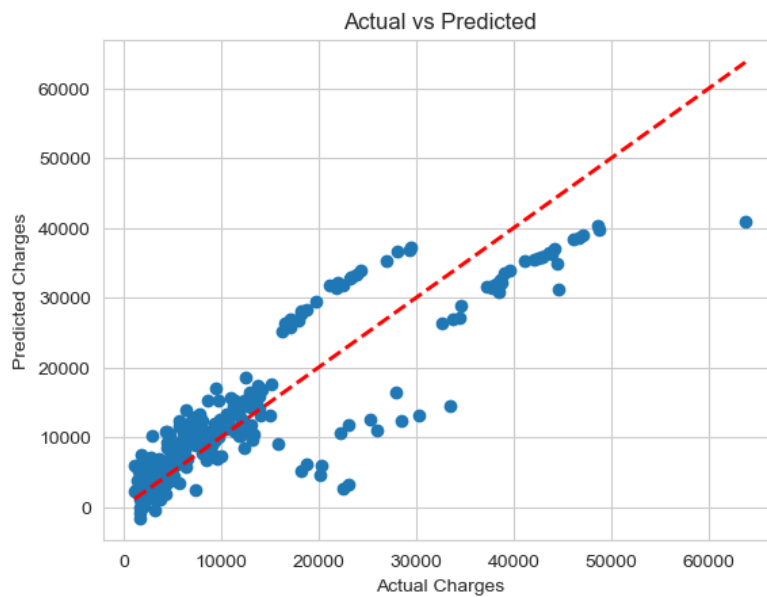


Figure 20: Actual insurance charges vs Predicted insurance charge

Retrain the model

SELECTING FEATURES

The model was retrained to ensure its relevance and accuracy. An interactive feature for `bmi_smoker` was created. The interaction represents the combined risk of these two factors. Being a smoker with a high BMI is far more harmful than one factor alone. The retrained model's five selected features were age, BMI, children, smoker_yes, and the BMI_smoker interaction. To calculate a baseline charge, the retrained Linear Regression model was setup with the `fit_intercept` parameter set to "True". The resulting model assigned specific coefficients to each feature, including the new interaction term. Using these parameters, the model achieved a convergence tolerance of $1e-06$, accurately representing the weighted influence of age, BMI, and smoking status on final insurance charges. The parameters (intercept and coefficients values) changed as shown on the below figure.

```
Intercept: -2684.314113083008
coefficients: [ 2.63517683e+02  9.34831546e+00  4.61809149e+02 -2.07995597e+04
 1.45633718e+03]
```

Figure 21:parameters for the retrained model

PREDICTIONS AND METRICS OF THE RETRAINED MODEL

The first ten predictions were generated, and they differ from the previous ten predictions, with a slight rise. By including the BMI * Smoker interaction, the model's explanatory power increased significantly, with the R square increasing from 0.78 to 0.86. This update significantly lowered the Mean Absolute Error by more than 40%, demonstrating that the model is now far more sensitive to the complex risk variables that underlie high medical costs. This indicates that for insurance data, a simple additive linear model is insufficient unless synergistic effects between attributes are considered. If you don't smoke, a high BMI raises your insurance premiums slightly due to general health concerns. However, if you are a smoker with a high BMI, your charges will significantly climb. The retrained model indicates that age, children, smoker_yes, and bmi_smoker_interaction features are highly statistically significant (p-values < 0.05). Notably, the bmi_smoker_interaction p-value demonstrates that the combination of smoking and a high BMI is a good predictor of medical charges. This statistical data supports the decision to eliminate inconsequential variables and focus on these main drivers, giving the medical aid system a strong and mathematically solid "sliding scale" for risk-based premium changes.

```
First 10 Predictions:
[10332.94377851  7082.95891861 32786.21247227 11063.59915066
 28307.6431837   7138.13900398 2532.99919368 14535.16552055
 4855.81339578 11881.68298982]
--- Improved Model Metrics ---
Mean Absolute Error (MAE): 2783.93354482586
Mean Squared Error (MSE): 21293425.767070208
Root Mean Squared Error (RMSE): 4614.480010474659
R2 Score: 0.8628431577993374

Statistically Significant Features (p < 0.05):
const                4.625773e-03
age                  3.536248e-104
children             2.159955e-04
smoker_yes           7.792892e-27
bmi_smoker_interaction 2.518034e-102
dtype: float64
```

Figure 22: predictions and metrics results

MODEL PREDICTED CHARGES VS ACTUAL VALUES

Actual values were compared with predicted values from the retrained model. The Actual vs. predicted figure supports the improved model's performance by demonstrating a good linear correlation between predicted values and actual ones. The tight clustering along the 45-degree identification line demonstrates that the model has accurately captured the various risk tiers. While some variance persists at the higher end of the charge spectrum, the addition of the interaction term has successfully reduced the systematic errors observed in the original linear regression model.

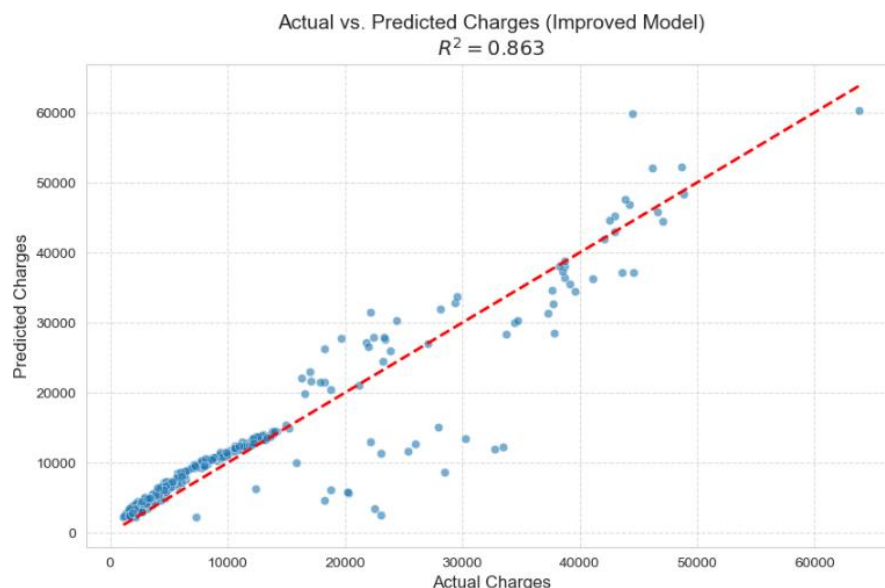


Figure 23: plot showing actual vs predicted charges

Conclusion

The study successfully developed a linear regression model that quantifies the association between personal factors and medical costs. The initial model demonstrated that smoking, age, children, and bmi are significant predictors; nevertheless, the finding of the BMI-Smoker interaction gave the most critical insight for the scheme's sliding scale. The retrained model even discards some of the chosen features (regions) because they have a very poor correlation with charges, and a p-value greater than 0.05.

We conclude that smoking has the greatest impact on costs, but a "sliding scale" of charges is most realistic when it accounts for overlapping risks rather than evaluating each item separately. This is because, for example, if you smoke and are 60 or older, or if you smoke have a high BMI, you are at a high risk of developing chronic diseases such as COPD, cancer, renal, diabetes, cardiovascular disease, stroke, and so on. The shift from the 8-feature baseline to the 5-feature enhanced model resulted in a considerable reduction in prediction error (MAE), giving the client a reliable, data-driven tool for making more precise and equitable premium adjustments.

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